Hunter Liles

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Computer Science and Computational Mathematics student seeking software engineering work, with focused knowledge of graphics and rendering.

Education

McNeese State University *Expected 2028*
Bachelor of Science in Computer Science and Computational Mathematics Lake Charles, LA

- Computer Science GPA: approximately 3.8/4.0

Research & Engineering Experience

Undergraduate Student Researcher *Current*
McNeese State University — Engineering & Computer Science Lake Charles, LA

- Develop a custom signed distance field rendering engine with another undergraduate researcher using C++, Raylib, and GLSL.
- Investigate mathematical geometry representations as an alternative to traditional triangle-based rendering.

Math & Science Tutor *Current*
MAST Department, McNeese State University Lake Charles, LA

- Help students understand mathematics and physics by translating difficult concepts into clear, workable steps.

Church IT *Current*
Iowa First Baptist Church Iowa, LA

- Maintain church computers and troubleshoot hardware and software for the media team.

Selected Projects

Blossom *C++, OpenGL*
2D game engine — active development
- Build a C++ game engine with OpenGL, GLFW, GLAD, and Dear ImGui.
- Implement custom vector and quaternion math libraries to support engine systems.

Signed Distance Field Rendering Engine *C++, GLSL*
Collaborative rendering research project — active development
- Create custom GLSL shaders for signed distance field functions and experiment with procedural geometry.
- Collaborate with another undergraduate researcher on a low-level rendering implementation.

Custom Profiler *Rust*
Collaborative developer tool — active development
- Develop a terminal profiler for logging frame times and CPU time to support application optimization.

Algorithmic Trading *Rust, Vulkan*
Collaborative paper-trading TUI — active development
- Build a terminal interface for testing and configuring paper-trading models with headless Vulkan.

Technical Skills

Languages: C, C++, Rust, Java, Python, HTML, CSS, SQL, GLSL, HLSL
Graphics & systems: OpenGL, Vulkan, graphics programming, physics simulation
Tools: Git/GitHub, Linux, SQL Servers, LaTeX, Raylib